



of life forms emerged. British naturalist Charles Darwin (1809–1882) explained the way in which species evolve with his theory of “natural selection.” He stated that simple structures, such as bacteria, can become complex organisms by adapting well to their environments. Not all plants and animals survive, because of predators, disease, or lack of food. Those that adapt by growing a furry winter coat or developing strong eyesight to avoid predators, for example, will be more likely to find a partner and mate. Their offspring inherit these adaptations, increasing their chance of survival.

EARTH'S TIMELINE

The evolutionary timeline starts 3.5 billion years ago, when carbon-containing molecules evolved into basic bacteria-like cells. About 450 million years ago, water-based life started to move onto land and develop into reptiles. The dinosaurs burst onto the scene about 230 million years ago, with the last ones dying out about 65 million years ago. Approximately five million years later, it was the turn of primates—the group of mammals

to which humans belong. However, humans are a young species compared to many primates and evolved about one million years ago. We're positively youthful in the grand scheme of things.

GOD'S GIFT

Creationism is the belief that the universe and all life forms were created by a god or gods. The first book of Christianity, Genesis, states that the universe was created in about 4000 BCE. Some religious enthusiasts agree, insisting that evolution cannot explain the rich diversity of life on Earth, but the special powers of a god can. While scientists believe that the universe is 13.7 billion years old, Christian creationists believe that it is only 6,000 years old. However, does a belief in evolution have to cancel out a belief that a god created the universe? Some people argue that God could have created the big bang and then instigated evolution. This allows the theories of creationism and evolution to exist side by side.

